

A Secure UAV Cooperative Communication Framework: Prospect Theory Based Approach

Liang Xie , Zhou Su , Qichao Xu , Nan Chen , Yixin Fan , and Abderrahim Benslimane 

Abstract—Unmanned Aerial Vehicles (UAVs) have attracted extensive attention from both industry and academia owing to their high mobility, line-of-sight (LoS) characteristics of air-to-ground (A2G) channels, and low cost. However, the broadcast nature of wireless transmission and the LoS characteristics of A2G channels are vulnerable to eavesdropping attacks, which leads to severe security issues. To enhance the security of UAV communication, we propose a framework that multiple UAVs cooperate to resist attacks (MURA). Specifically, we first introduce an efficient incentive scheme based on the coalition game to motivate UAVs to participate in a coalition. By forming a grand coalition, we demonstrate that each individual UAV can maximize its utility. Additionally, we propose a secure UAV communication scheme designed to counter eavesdropping attacks. Two distinct types of scenarios are considered in UAV communication. In a completely rational scenario, in which participants make decisions aiming to maximize their utility, we utilize the Stackelberg game to model the interactions between UAVs and attacker. The existence and uniqueness of the equilibrium solution are proved, and the equilibrium solution is obtained. In the context of an imperfectly rational scenario, we employ the prospect theory (PT) to effectively capture the underlying rationality exhibited by the players. The PT valuations of the players, i.e., UAV and attacker, are deduced in detail. Meanwhile, the convergence of the PT valuations of UAV and attacker is proved. Finally, extensive simulation results show that the proposed scheme can effectively improve the utility of legal UAVs and ensure the security of the UAV networks compared with benchmarks.

Index Terms—Unmanned aerial vehicles, eavesdropping attack, prospect theory, game theory.

I. INTRODUCTION

WITH the rapid development of technologies, such as electronics and wireless communication, unmanned aerial

vehicles (UAVs) are widely used in military and civilian fields owing to their high mobility, low cost, line-of-sight (LoS) characteristics with ground node channels, and on-demand deployment [1], [2], [3], [4]. For example, UAVs can be utilized as aerial base stations for strong LoS channel links with ground nodes, which can significantly enhance the transmission performance and increase the capacity and coverage of wireless networks [5], [6]. Specifically, UAVs can crucially establish connections with ground users in regions where terrestrial infrastructure is unavailable or compromised, or facilitate rapid business recovery after infrastructure is destroyed in natural disasters [7], [8], [9]. In addition, UAVs can be used as relays to transmit information from the source node to the distant destination node [10]. Furthermore, UAV deployments allow for more flexible network architectures and lower deployment costs than traditional base stations [11], [12].

Despite the advantages of UAVs in providing flexible, scalable, and cost-efficient communication, it is necessary to consider the security performance of the wireless transmission between UAVs and ground nodes due to the broadcasting characteristics of the wireless channel [13]. The eavesdropping attack is the most common and most concerned type of attack in UAV networks [14]. On the other hand, the eavesdropping attack greatly threatens the security of UAV networks. Specifically, eavesdroppers can easily access the secret information transmitted between UAVs and ground nodes due to the strong LoS characteristics of UAV communication, thus threatening the security of UAV networks [15]. Recently, the physical layer security has been studied to address the security issues of UAV networks [16], [17], [18]. To ensure secure communication between UAVs and ground users without sharing passwords, signal processing and channel coding are utilized in physical layer security [19], [20].

However, in the existing physical layer security studies, jamming UAVs are considered selfless, i.e., making decisions regardless of their energy consumption and profits [21]. In addition, the existing research lacks the appropriate defense schemes, which will make the physical layer security ineffective [22]. Furthermore, as the attackers' computing power increases dramatically, a single UAV may not be able to effectively resist malicious attacks due to limitations in computing power and defense capabilities [11], [23].

To tackle the above challenges, we propose a framework in which multiple UAVs cooperate to resist attacks (MURA) to enhance the security of UAV networks by employing physical layer security technology. Specifically, in the proposed framework, legal UAVs that operate in full duplex mode are equipped with

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Liang Xie and Zhou Su are with the School of Cyber Science, Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: zhousu@ieee.org).

Qichao Xu is with the School of Mechatronic Engineering, Automation, Shanghai University, Shanghai 200444, China.

Nan Chen is with the Department of Electrical, Computer Engineering, Tennessee Tech University, Cookeville, TN 38505 USA.

Yixin Fan is with the School of Communications Engineering, Xidian University, Xi'an 710126, China.

Abderrahim Benslimane is with the Computer Science, Engineering, University of Avignon, 84029 Avignon, France.

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two antennas, where one receiving antenna is used to receive data from ground users and one transmitting antenna is used to transmit jamming signals to the eavesdropping UAV [24]. Nearby UAVs can form a coalition to resist attacks. In the coalition, these UAVs are divided into the leader UAV and follower UAVs based on their defense capabilities. The leader UAV is the UAV with the best defense capabilities. Follower UAVs can join the coalition to resist attacks or form a separate coalition to resist attacks. In the case of joining the coalition, the follower UAV has access to some of the defensive resources of the leader UAV and other follower UAVs. Besides, some of its defensive resources are used to assist other follower UAVs in the coalition. Conversely, when legal UAVs decide not to join the coalition, they will resist attacks independently. The attacker is an active eavesdropper capable of threatening the security of the UAV networks by varying the attack intensity.

In the proposed MURA framework, the purpose of the UAV is to maximize its benefit while securing network communication, where the benefit of the UAV is positively related to the data transmission rate of the UAV defined in the communication model. Specifically, we first propose an incentive scheme based on the coalition game to stimulate legal UAVs to join the coalition to collaboratively resist eavesdropping attacks, thus improving the security of UAV networks. Afterwards, a secure UAV communication scheme is proposed to resist eavesdropping attacks, where the interaction between leader UAV and attacker is formulated as a Stackelberg game. Specifically, The leader UAV is the game-leader to determine the interference power strategy, while the attacker is the game-follower to make the attack intensity strategy. Furthermore, in a more practical case, we apply prospect theory (PT) to the above game to capture the bounded rationality factors of the players. The main contributions are summarized as follows:

- We propose a framework that multiple UAVs cooperate in defending against attackers, which can address the security issues incurred by the resource limitation of UAVs.
- To motivate legal UAVs to collaboratively resist eavesdropping attacks, an incentive scheme based on the coalition game is proposed. In addition, a secure UAV communication scheme is proposed to enhance the security of UAV networks, where the optimal strategy for leader UAV and attacker can be obtained. The existence and uniqueness of the Stackelberg equilibrium are proved and obtained.
- To model the realistic decision-making process of leader UAV and attacker under conditions of risk, we apply PT to the Stackelberg game to theoretically analyze the impact of limited rationality of players. Besides, we derive in detail the PT valuations of leader UAV and attacker while proving their convergence.
- The effectiveness of the proposed scheme is evaluated through extensive simulations. We show that the proposed scheme can efficiently improve the utilities of legal UAVs, while ensuring the security of the UAV network, compared with benchmarks.

The remainder of this paper is organized as follows. Related work is reviewed in Section II. Section III presents the system model. The incentive scheme for legal UAVs based on the coalition game is proposed in Section IV. In Section V,

we formulate the secure UAV communication scheme. The secure UAV communication scheme based on PT is analyzed in Section VI. Extensive simulations are conducted to evaluate the performance of the proposed scheme in Section VII. Finally, we give the conclusion in Section VIII.

II. RELATED WORK

Extensive studies have been conducted to address the security issues in UAV networks. Physical layer security can ensure secure communication between UAVs and ground users by using signal processing technology and channel coding. Zhang et al. [18] proposed a new problem of maximizing the average secrecy rate by combining UAVs with physical layer security, in which UAVs resisted eavesdropping attacks by emitting jamming signals. Zhou et al. [24] studied a secure mobile edge computing system for UAV, where a low-complexity iterative algorithm was designed to minimize the waiting time and maximize the secrecy rate. Liu et al. [25] presented a secure UAV network that leverages physical layer security to disrupt eavesdropping channels, where an iterative suboptimal algorithm was designed to maximize the secrecy of the system. Chen et al. [26] developed a UAV network architecture that incorporates physical layer security, where John Boyd's OODA loop-based secure mechanism was devised to protect data from interference.

Intrusion detection is a reliable technology to protect UAV networks from attacks because it can detect both internal and external attacks with a high degree of accuracy. Sedjelmaci et al. [27] designed a novel intrusion detection and response scheme to detect the attack type according to the behavior of the UAV. Mitchell et al. [28] proposed an intrusion detection system to audit the UAVs in the system to determine whether their operational status is normal or under attack. Alpcan et al. [29] proposed a secure scheme to detect the most valuable or vulnerable points in the internet of vehicles, where a secure game model was proposed to model the interactions between the attacker and UAV. Sedjelmaci et al. [30] constructed an intrusion detection system to protect the ground network composed of sensors and ground vehicles, where the Bayesian game was exploited to address the problem of detection rate and false alarm rate to achieve accurate detection of attacks at a lower cost.

While the existing works extensively explore the security of UAV networks, a more realistic scenario in which UAVs with bounded rationality should be considered. Sanjab et al. [31] proposed a novel mathematical framework to model and analyze the security of the UAV interactive system, where cumulative prospect theory was used to capture the subjective evaluation of participants and the subjective perception of danger. Xiao et al. [32] proposed an intelligent attack game based on prospect theory, where a deep Q learning algorithm was designed to obtain the equilibrium solution of the dynamic game. Sanjab et al. [33] researched the security of UAV delivery systems, where prospect theory was formulated to prove that the subjective decision-making process of participants will lead to longer interaction time.

While existing works address the security issues in single UAV networks from various aspects, few of them consider a network with multiple UAVs. Further, the design of incentive

TABLE I
SUMMARY OF NOTATIONS

Notations	Descriptions
$\mathcal{K}$	The set of ground nodes in the UAV networks.
$\mathcal{I}$	The set of legal UAVs in the UAV networks.
$\mathcal{J}$	The set of eavesdropping UAVs in the UAV networks.
p_i	the interference power of legal UAV i .
R_u	The radius of the legal UAV airspace Ω_u .
R_e	The radius of the eavesdropping UAV airspace Ω_e .
λ	The attack intensity of attacker.
h	The channel power gain at the reference distance $d_0 = 1$.
H_i	The altitude of legal UAV i .
H_j	The altitude of eavesdropping UAVs j .
n	The number of eavesdropping UAVs selected by the attacker.
γ^+, γ^-	The rational parameters of player in PT.
p_i	The transmit power of legal UAV i .
$p_a(n)$	The attack probability of the attacker.
μ	The mean value of the uniform distribution of eavesdropping UAVs in the airspace Ω_e .
∇	The area of airspace Ω_e .
p_s	The probability of the attacker successfully attacking UAV networks.
δ	The proportion of defensive capabilities that legal UAVs need to share when joining the coalition.
$r_{i,k}$	The data transmission rate between legal UAV i and ground node k .
$r_{j,k}$	The data transmission rate between eavesdropping UAV j and ground node k .
P	The interference power of the leader UAV.
M	The total number of legal UAVs.
N	The total number of eavesdropping UAVs.
β	The trust coefficient of legal UAV for other legal UAVs except the leader UAV.
θ	The energy consumption factor of leader UAV.
ζ	The weigh parameter of attacker.
ξ	The degree of risk appetite of players for profits.
ϖ	The degree of risk appetite of players for loss.
$\tilde{\lambda}_S$	The loss aversion coefficient of each player in PT.
R_S	The reference point of player in PT.

schemes for UAVs to resist malicious attacks is rarely discussed. Different from the above research, we consider the possibility of combining multiple UAVs into a coalition with high computation resources to resist malicious attacks. Moreover, we utilize PT to capture the subjective evaluation of participants under incomplete rationality.

III. SYSTEM MODEL

In this section, we introduce the system model, including the network model, mobility model, channel model, and attack model. The notations utilized in this paper are summarized in Table I.

A. Network Model

As illustrated in Fig. 1, the network model consists of the following three entities: legal UAVs, an attacker, and ground nodes.

- **Ground nodes:** Ground nodes are randomly distributed on the ground with the ability of storage and calculation. Let $\mathcal{K} \triangleq \{k|1, 2, \dots, K\}$ denotes a set of ground nodes in the UAV networks, where ground node k is in charge of transmitting data or initiating content requests to legal UAVs, such as traffic conditions, over wireless networks. The positions of the ground nodes are represented by a three-dimensional Cartesian coordinate system. Thus, the coordinates of ground node k are (x_k, y_k) , since the ground

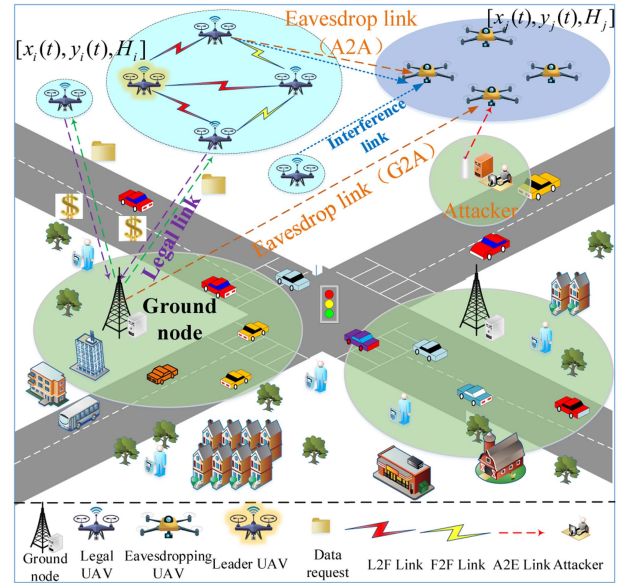


Fig. 1. Illustration of MURA framework.

nodes are deployed on the ground, the height is zero and therefore omitted in the coordinates.

- **Legal UAVs:** Let $\mathcal{I} \triangleq \{i|1, 2, \dots, I\}$ denotes a set of legal UAVs, where legal UAVs are evenly distributed in a circular airspace Ω_u with radius R_u . In addition, each legal UAV flies in the airspace Ω_u at the average flight speed $\bar{v}_1$

and establishes a LoS link with a ground node to transmit confidential information. When the data transmission rate is slow due to the distance between legal UAV and ground node, the legal UAV will increase its transmit power or fly closer to the ground node. In addition, legal UAVs are divided into two categories based on their defensive capability: the leader UAV and follower UAVs. The leader UAV is the legal UAV with the best defense capability and is completely trusted by other legal UAVs. The follower UAVs are all legal UAVs except the leader UAV and have different defensive capabilities. Based on [32], [34], the defensive capability of the legal UAV is measured by its interference power. Specifically, the higher the interference power of the legal UAV, the greater its ability to affect eavesdropping attacks and thus the higher the security of the UAV networks. In addition, experiments in [24] demonstrate that the defensive capability of the legal UAV can be mathematically modeled as a concave function in relation to the interference power. Accordingly, the natural logarithm function is employed to formulate the defense capability of the legal UAV, which can be calculated by

$$f(p_i) = \log(1 + p_i), \quad (1)$$

where p_i is the interference power of legal UAV i . Assume that the interference power of the leader UAV is much greater than that of other legal UAVs.

- **Attacker:** The purpose of the attacker is to eavesdrop on communications between legal UAVs and ground nodes by launching eavesdropping UAVs. Let $\mathcal{J} \triangleq \{j|1, 2, \dots, J\}$ denotes a set of eavesdropping UAVs controlled by the attacker. The coordinates of the attacker are denoted as $(x_{attacker}, y_{attacker})$, since he/she is located on the ground as shown in Fig. 1. Similar to [14], [18], [35], we assume the attacker is an active eavesdropper capable of threatening the security of the UAV network by varying the attack intensity, which is measured by the number of eavesdropping UAVs selected by the attacker to launch the attack. Therefore, by using a camera or synthetic aperture radar (SAR) equipped on the UAV, the real-time locations of potential eavesdropping UAVs can be physically detected and traced by the leader UAV, while the number of eavesdropping UAVs can be roughly detected [18]. In the case of eavesdropping UAVs that passively receive the signal from the transmitter, which is outside the scope of this paper, we may study this aspect in the future. We assume that eavesdropping UAVs are located in a circular airspace Ω_e with radius R_a , where the distance between airspace Ω_e and airspace Ω_u is d_u . In addition, each eavesdropping UAV flies in the airspace Ω_e at the average flight speed $\bar{V}_2$ to eavesdrop on the communication between the legal UAV and the ground node. Furthermore, when the interference power of the legal UAV is high, the eavesdropping UAV will fly closer to the legal UAV to perform eavesdropping attacks. When the interference power of the legal UAV is low, the eavesdropping UAV will fly away from the legal UAV to hide.

B. Mobility Model

Let $\mathcal{T} = \{1, \dots, t, \dots, T\}$ be the finite time horizon. By dividing the finite time horizon into T time slot, where each time slot length is ϱ . Since ϱ can be adequately small, the position of the UAV can be approximately fixed in each time slot. As such, the horizontal locations of legal UAV i and eavesdropping UAV j at time slot t are expressed as $\Theta_i(t) = [x_i(t), y_i(t)]$ and $\Theta_j(t) = [x_j(t), y_j(t)]$, respectively. The altitude of legal UAV i is fixed at H_i , which meets the basic obstacle avoidance requirements [36], [37]. Similarly, the altitude of eavesdropping UAV j is fixed at H_j . In the working area, the flying trajectory of each UAV is predefined, where the initial and termination positions of legal UAV i are denoted separately as $\Theta_i(1)$ and $\Theta_i(T)$ regardless of the altitude. Likewise, the initial and termination positions of eavesdropping UAV j are denoted separately as $\Theta_j(1)$ and $\Theta_j(T)$. Apparently, the flying trajectories of legal UAV i and eavesdropping UAV j within time horizon T can be formulated separately as

$$\begin{cases} \Theta_i = \{\Theta_i(1), \dots, \Theta_i(t), \dots, \Theta_i(T)\}, \\ \text{s.t. } \|\Theta_i(t+1) - \Theta_i(t)\| \leq \varrho \mathcal{V}_i^{\max}, 1 \leq t \leq T, \end{cases} \quad (2)$$

$$\begin{cases} \Theta_j = \{\Theta_j(1), \dots, \Theta_j(t), \dots, \Theta_j(T)\}, \\ \text{s.t. } \|\Theta_j(t+1) - \Theta_j(t)\| \leq \varrho \mathcal{V}_j^{\max}, 1 \leq t \leq T, \end{cases} \quad (3)$$

where $\mathcal{V}_i^{\max}$ and $\mathcal{V}_j^{\max}$ are the maximum flight speeds of legal UAV i and eavesdropping UAV j , respectively. Based on the mobility model, the number of UAVs in the airspace is time-varying. Therefore, the set of UAVs in the airspace Ω_u at time slot t is denoted as $\mathcal{I}(t) = \{i|1, 2, \dots, I(t)\}$, where $I(t)$ is the number of legal UAVs in the airspace Ω_u at time slot t .

C. Channel Model

The wireless channel model of UAV-to-ground/ground-to-UAV link is a hot topic of current research. Most of the existing models can be divided into three categories: free-space LoS models, models in which channel parameters depend on altitude/angle, and probabilistic LoS channel models. The recent research in [18] shows that the LoS model provides a good approximation for the actual UAV-to-ground communication. Inspired by this, we employ the LoS communication model for UAV-to-ground communication. To meet the basic obstacle avoidance requirements and keep UAVs flying without hourly ascending and descending due to terrain or building blockage in the execution of the sensing tasks, we set the flight altitude of legal UAV i to H_i based on the type of terrain (i.e., flight altitude H_i greater than the height of the largest obstacle in the task area $H_s^{\max}$, which can be measured prior to the task execution). For example, in rural areas, the UAV flight altitude is set to 100 m to avoid obstacles. In this case, the air-to-ground link between the UAV and ground users is guaranteed to be a LoS channel [38]. In addition, the altitude of the UAV needs to be less than its maximum communication range $H_c^{\max}$, which can be measured in advance using signal strength measuring instruments. Thus, the channel gain between ground node k and legal UAV i at time slot t , denoted as $g_{i,k}(t)$, follows the free-space path loss model,

which is defined as

$$g_{i,k}(t) = \frac{\bar{h}}{H_i^2 + (x_i(t) - x_k(t))^2 + (y_i(t) - y_k(t))^2}, \quad (4)$$

where $\bar{h}$ represents the channel power gain at the reference distance $d_0 = 1$, which depends on the carrier frequency ρ and the antenna gain g_t/g_r of the transmitter/receiver, is calculated by $\bar{h} = g_t g_r (\frac{c}{4\pi\rho d_0})^2$. Here, c is the speed of light.

Similarly, the LoS channel gain between eavesdropping UAV j and ground node k at time slot t , denoted as $g_{j,k}(t)$, is updated by

$$g_{j,k}(t) = \frac{\bar{h}}{H_j^2 + (x_j(t) - x_k(t))^2 + (y_j(t) - y_k(t))^2}. \quad (5)$$

The frequency division multiple access (FDMA) protocol is used to achieve bandwidth sharing among UAVs when transmitting data with ground nodes, thus the interference between UAVs can be eliminated [39]. In addition, in order to avoid the self-interference of UAVs, we eliminate the self-interference of legal UAVs by separating their transmitting antennas and interfering antennas at a distance and using advanced analog and digital self-interference cancellation methods [40], [41], [42]. Thus, if legal UAV i is connected to ground node k , the data transmission rate $r_{i,k}(t)$ at time slot t is defined as

$$r_{i,k}(t) = \mathcal{B}_i \log_2 \left(1 + \frac{g_{i,k}(t) \mathbf{p}_i}{\sigma^2} \right), \quad (6)$$

where σ^2 represents the power of Gaussian white noise, $\mathbf{p}_i$ is the transmit power of legal UAV i . $\mathcal{B}_i$ is the assigned bandwidth to UAV i .

Similarly, the data transmission rate between eavesdropping UAV j and ground node k at time slot t , denoted as $r_{j,k}(t)$, is defined as

$$r_{j,k}(t) = \mathcal{B}_j \log_2 \left(1 + \frac{g_{j,k}(t) \mathbf{p}_i}{\sigma^2} \right), \quad (7)$$

where $\mathcal{B}_j$ is the assigned bandwidth to UAV j .

The secrecy rate is the primary design metric for physical-layer security, which is expressed as the ratio of confidential information that can be reliably transmitted without being eavesdropped. The secrecy rate is defined as the available transmission rate between the legal UAV and the ground node minus the available transmission rate between the eavesdropping UAV and the ground node. Therefore, the average secrecy rate from legal UAV i to ground node k is given by

$$R_{\text{sec}}^k = \frac{1}{T} \sum_{t=1}^T [r_{i,k}(t) - r_{j,k}(t)]^+, \quad (8)$$

where $[x]^+ \triangleq \max(x, 0)$.

D. Attack Model

We consider a framework consisting of multiple legal UAVs and an attacker, where the attacker can manipulate eavesdropping UAVs by using smart devices to eavesdrop on the information transmitted by the legal UAV. Specifically, the attacker

first chooses the number of eavesdropping UAVs, and then chooses the optimal attack intensity based on the strategy of the leader UAV. In addition, we consider that the leader UAV will resist eavesdropping attacks by emitting jamming signal, so the attacker is not guaranteed to succeed in every attack. We assume that the number of eavesdropping UAVs follows the Poisson process [20]. Empirically, when the number of eavesdropping UAVs in the airspace Ω_e increases, the attacker's probability of a successful attack increases, and thus his or her gains increase. However, the cost of the attacker is positively related to the number of eavesdropping UAVs used to launch the attack. Given that the attacker is a rational entity, the goal is to maximize its utility. In contrast to the probability of attack, which is positively correlated with the number of eavesdropping UAVs, the attacker is guided by a concave function in choosing the probability of launching an attack in order to maximize its utility. Therefore, the probability of the attacker launching an attack increases and then decreases as the number of eavesdropping UAVs increases. This intrinsic relationship satisfies the Poisson process. Therefore, the probability of the number of eavesdropping UAVs in airspace Ω_e is quantified as the probability of the attacker launching attack, denoted as $p_a(n)$, which is calculated by

$$p_a(n) = e^{-\mu} \frac{\mu^n}{n!}, \quad (9)$$

where $\mu = \lambda \mathbb{V}$ is the mean value of the uniform distribution of eavesdropping UAVs in the airspace Ω_e . λ is the attack intensity of the attacker, defined as the number of eavesdropping attacks per unit of time. $\mathbb{V} = \pi R_a^2$ is the area of airspace Ω_e . n represents the number of eavesdropping UAVs selected by the attacker in airspace Ω_e . Let p_s be the probability of a successful attack. The probability of the attacker successfully attacking the communication between legal UAVs and ground nodes is $p_a(n)p_s$. A successful attack will bring benefits to the attacker and severe security issues, such as the loss of confidential information and the leakage of privacy.

IV. INCENTIVE SCHEME FOR LEGAL UAVS BASED ON THE COALITION GAME

As the number of legal UAVs in the coalition increases, the communication between legal UAVs and ground nodes is more reliable and secure. As such, we propose an incentive scheme based on the coalition game to motivate legal UAVs to join the coalition. The leader UAV will assist legal UAVs that join the coalition in defending against attackers by sharing power and resources. We utilize the coalition game $G[Q, \mathbb{S}]$ to characterize the interactions among legal UAVs. Q is the set of players in the game, and $\mathbb{S}$ is an arbitrary subset of Q , expressed as a coalition. The strategy of Q is whether to join the coalition. Q_i represents legal UAV i . After selecting the strategy, Q_i then transmits data to the ground nodes via radio. Specifically, in the case of joining the coalition, the legal UAV can obtain part of the leader's defense capabilities depending on the defense status of other UAVs in the coalition, while some of its defense capabilities are used to assist other legal UAVs in the coalition. The physical meaning of the defense capacity refers to a UAV's ability to withstand eavesdropping attacks. Note that

the defensive capability of the legal UAV can be mathematically modeled as a concave function in relation to the interference power. Therefore, legal UAVs within a coalition are able to effectively exchange information about their defense tasks and divide their defense capabilities equally among other legal UAVs through a defense capability distribution strategy. In the case that the legal UAV does not join the coalition, the legal UAV defends against attackers independently. Note that the purpose of legal UAVs is to maximize their utility. To maximize the utility of legal UAVs, the coalition game is utilized to analyze and model the legal UAVs group.

Due to the time-varying decision-making process of legal UAVs and the mobility of both legal UAVs and eavesdropping UAVs, the decision-making can be done in each time slot. We choose to examine the decision-making process within a single time slot rather than considering the entire time horizon. Therefore, for the sake of illustration, we omit the “(t)” in each of the following symbols, e.g., $r_{i,k}(t)$ is converted to $r_{i,k}$. Among legal UAVs, some may be sensitive to the benefits of data transmission, while others may be sensitive to the losses caused by attacks. Therefore, the utility of legal UAV Q_i is defined as

$$\tilde{\Omega}(i) = (1 - p_s p_a(n)) r_{i,k} F(p_i) |_{\tilde{\omega}}, \quad (10)$$

where $\tilde{\omega} = \{0, 1\}$ represents two defense modes of UAVs, i.e., whether to join the coalition. If UAV Q_i chooses to join the coalition, $\tilde{\omega} = 1$. Otherwise, $\tilde{\omega} = 0$. $F(p_i) |_{\tilde{\omega}}$ is the defense capability of UAV Q_i in different defense modes, defined as

$$F(p_i) |_{\tilde{\omega}} = \begin{cases} \log(1 + p_i), & \tilde{\omega} = 0, \\ (1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) \\ + \delta \beta \frac{1}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'}), & \tilde{\omega} = 1. \end{cases} \quad (11)$$

where P is the interference power of the leader UAV. M is the total number of legal UAVs. δ is the proportion of defensive capabilities that legal UAVs need to pay when joining the coalition. $\beta \in [0, 1]$ is the trust coefficient of the legal UAV for other legal UAVs except the leader UAV, which depends on their previous interactions. Specifically, the higher the number of positive interactions (i.e., willingness to form coalitions together to collaboratively defend against eavesdropping attacks) that occur among legal UAVs, the higher the level of trust between them. We assume that the leader UAV is honest and that all legal UAVs trust the leader UAV completely. Therefore, the level of trust of the leader UAV is 1.

Substituting (11) into (10), the utility of legal UAV Q_i can be rewritten as

$$\tilde{\Omega}(i) = \begin{cases} (1 - p_s p_a(n)) r_{i,k} \log(1 + p_i), & \tilde{\omega} = 0, \\ (1 - p_s p_a(n)) r_{i,k} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) \\ + \delta \beta \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})], & \tilde{\omega} = 1. \end{cases} \quad (12)$$

The interactions among legal UAVs are formulated as a coalition game with transferable benefits, in which each legal UAV chooses the optimal access link by forming a coalition with other legal UAVs. When all legal UAVs in the coalition acquire

the defense resources they need, the cooperation of legal UAVs within the coalition is complete. For legal UAV Q_i , we have $\mathcal{Z}_i = 1$, if

- No other legal UAVs request defense resources.
- No other legal UAVs can provide defense resources.
- UAV Q_i cannot provide other legal UAVs defense resources.

If $\mathcal{Z}_i = 1$, UAV Q_i will form a coalition by itself and defend against the attacker independently.

Definition 1: For a coalition game with transferable benefits, the coalition is super-additive, if two coalitions $\mathcal{S}_q$ and $\mathcal{S}_{q'}$ ($\mathcal{S}_q, \mathcal{S}_{q'} \subset \mathbb{S}, \mathcal{S}_q \cap \mathcal{S}_{q'} = \emptyset$) satisfy $\tilde{\Omega}(\mathcal{S}_q) + \tilde{\Omega}(\mathcal{S}_{q'}) \leq \tilde{\Omega}(\mathcal{S}_q \cup \mathcal{S}_{q'})$.

It can be seen that if a coalition game is super-additive, the grand coalition formed by all legal UAVs can maximize the utility of each legal UAV. In other words, the grand coalition can achieve the maximal utility of legal UAVs compared with other coalition structures.

Theorem 1: The proposed transferable benefits coalition game is super-additive.

Proof: Consider a simple case where there are two coalitions $\mathcal{S}_1$ and $\mathcal{S}_2$, $\mathcal{S}_1 = \{i\}$, $\mathcal{S}_2 = \{h\}$. $\mathcal{S}_1$ and $\mathcal{S}_2$ can be combined into a new coalition, denoted as $\mathcal{S}_3$, $\mathcal{S}_3 = \{i, h\}$. The coalition game is super-additive, if satisfy

$$\tilde{\Omega}(\mathcal{S}_q) + \tilde{\Omega}(\mathcal{S}_{q'}) - \tilde{\Omega}(\mathcal{S}_q \cup \mathcal{S}_{q'}) \leq 0. \quad (13)$$

Substituting $\tilde{\Omega}(\mathcal{S}_q)$, $\tilde{\Omega}(\mathcal{S}_{q'})$, and $\tilde{\Omega}(\mathcal{S}_q \cup \mathcal{S}_{q'})$ into (12), we can obtain

$$\begin{aligned} & \tilde{\Omega}(\mathcal{S}_q) + \tilde{\Omega}(\mathcal{S}_{q'}) - \tilde{\Omega}(\mathcal{S}_q \cup \mathcal{S}_{q'}) \\ &= (1 - p_s p_a(n)) r_{i,k} \log \left[\frac{(1 + p_i)^{\delta(1-\beta)} (1 + p_h)^{\delta(1-\beta)}}{(1 + P - p_i)(1 + P - p_h)} \right]. \end{aligned} \quad (14)$$

Since the interference power of the leader UAV is much greater than that of other legal UAVs in the coalition, that is, $P \gg p_i$. If $P > p_i + p_j$ holds, $(1 + P - p_i)(1 + P - p_h) \geq (1 + p_i)(1 + p_h)$, so as to $(1 + P - p_i)(1 + P - p_h) > (1 + p_i)^{\delta(1-\beta)} (1 + p_h)^{\delta(1-\beta)}$. Therefore, we can obtain the inequality shown in (13). Theorem 1 is proved. ■

Remark: In Theorem 1, we consider a simple case in which two nodes are combined as a new coalition, where the leader UAV already exists in the new coalition. In our framework, the leader UAV is selected in advance based on the magnitude of interference power of each legal UAV and the level of trust based on historical interactions in each time slot. Moreover, we assume that the leader UAV is the initiator of the coalition, i.e., the first to participate in the coalition. Other legal UAVs choose whether or not to join the grand coalition that exists for the leader UAV. Based on Theorem 1, the proposed transferable benefits coalition game is super-additive, that is, joining the coalition could maximize the utility of legal UAVs. Therefore, it can motivate legal UAVs to join the coalition.

V. SECURE UAV COMMUNICATION SCHEME

In this section, we present the detailed design of a secure UAV communication scheme to resist the eavesdropping attack. Specifically, the problem formulation is first proposed, where the interaction between the leader UAV and the attacker is formulated as a Stackelberg game. Then, the utility of the leader UAV and the attacker is studied. Finally, the optimal strategy for the leader UAV and the attacker is derived in detail.

A. Problem Formulation

After stimulating UAVs to join the coalition by using an incentive scheme based on the coalition game, there still exists security issues due to the broadcasting characteristics of wireless channels. To resist the eavesdropping attack, the leader-follower hierarchical Stackelberg game is exploited to model the interaction between the leader UAV and the attacker. The leader UAV is the game-leader to determine the interference power strategy, while the attacker is the game-follower to make the attack intensity strategy. In this game, the set of players is $S = \{S_U, S_A\}$. S_U is the leader UAV, and S_A represents the attacker. Specifically, at this point all legal UAVs are considered as a whole, the leader UAV first chooses the interference power P based on the trade-off between energy consumption and security. Next, given the interference power P chosen by the UAV, the attacker chooses the attack intensity to eavesdrop on the communications between the legal UAVs and the ground nodes. Therefore, a leader-follower (i.e., the leader UAV is the leader and the attacker is the follower) hierarchical game model is adopted.

As the leader of the game, leader UAV chooses the interference power P in the first stage, where $P \leq P_{\max}$. $P_{\max}$ is the maximum interference power depending on the energy of the leader UAV. Given the interference power P of the leader UAV, the attacker chooses the attack intensity λ , where $0 \leq \lambda \leq \lambda_{\max}$. $\lambda_{\max}$ represents the upper limit of attack intensity.

B. Utility Function

1) The utility of UAV group

The secrecy rate is an influential factor in the utility of legal UAVs. Specifically, the utility of the legal UAV increases as the secrecy rate increases. However, an increase in the utility of the legal UAV does not represent an increase in the secrecy rate. This is because the purpose of the legal UAV is to maximize its benefits while securing network communication. Therefore, we define the utility function to quantitatively characterize the preferences and benefits of legal UAVs under different strategies and thus predict their decisions. Thus, the utility of legal UAVs depends not only on the secrecy rate but also on the cost of defending against eavesdropping attacks. Specifically, the utility of UAV group is composed of the profits obtained by transmitting data with ground nodes, the loss cost caused by the successful attack, and the energy cost of resisting the attacker. As such, the utility of UAV group is defined by

$$U_{S_U}(P, \lambda) = G(P, \lambda) - L(P, \lambda), \quad (15)$$

where $G(P, \lambda)$ is the profits obtained by transmitting data with ground nodes. $L(P, \lambda)$ represents the cost function, which consists of two components: the cost of loss caused by a successful attack and the cost of energy consumption.

The probability of successful data transmission is positively related to the security of UAV networks. As such, the profit of UAV group is calculated by

$$G(P, \lambda) = (1 - p_s p_a(n)) \sum_{i \in \mathcal{I}(t)} r_{i,k} \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log \times (1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})], \quad (16)$$

The cost function is related to the attack intensity and the energy consumption of legal UAVs. If the attack intensity is high, the legal UAV needs to consume more energy to defend against the attacker and incur more loss costs. As such, the cost of UAV group is defined as

$$L(P, \lambda) = \chi p_s p_a(n) \sum_{j \in \mathcal{J}} r_{j,k} + e^{\theta P}, \quad (17)$$

where θ is the parameter for weighing cost against energy consumption of legal UAVs. χ is the parameter for weighing cost against attack intensity.

Substituting (16) and (17) into (15), the utility of UAV group can be rewritten as

$$U_{S_U}(P, \lambda) = (1 - p_s p_a(n)) \sum_{i \in \mathcal{I}(t)} r_{i,k} \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \times \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})] - \chi p_s p_a(n) \sum_{j \in \mathcal{J}} r_{j,k} - e^{\theta P}. \quad (18)$$

2) The utility of attacker

The utility of attacker is based on the benefits of data transmission between eavesdropping UAVs and ground nodes and the cost of the attack, which can be expressed by

$$U_{S_A}(P, \lambda) = \Gamma(\lambda) - \Lambda(P, \lambda), \quad (19)$$

where $\Gamma(\lambda)$ is the benefit function of attacking the data transmission between legal UAVs and ground nodes. $\Lambda(P, \lambda)$ represents the cost function of the attacker to launch eavesdropping attack. If the attacker can successfully attack the data transmission between legal UAVs and ground nodes, it will obtain large profits. As such, the benefit function of attacker is calculated by

$$\Gamma(\lambda) = \zeta p_s p_a \sum_{j \in \mathcal{J}} r_{j,k}, \quad (20)$$

where ζ is the weighting parameter.

The cost function is related to the attack intensity and the defense capabilities of the UAV group. In other words, to achieve a higher probability of a successful attack, the attacker needs to choose more eavesdropping UAVs, which leads to an increase in

costs. In addition, the defensive capability of the UAV group is negatively related to the probability of a successful attack, so the attack cost increases with the defensive capability of the UAV group. Therefore, the cost of attacker is defined as

$$\Lambda(P, \lambda) = p_s p_a \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})]. \quad (21)$$

Substituting (20) and (21) into (19), the utility function of attacker can be rewritten as

$$U_{S_A}(P, \lambda) = \zeta p_s p_a \sum_{j \in \mathcal{J}} r_{j,k} - p_s p_a \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})]. \quad (22)$$

According to the above analysis, the optimization problems for the leader UAV and the attacker can be formulated separately. The goal of the leader UAV is to determine the optimal strategy to maximize its utility under the power constraint. As such, the optimization problem for the leader UAV can be formulated as *Problem 1*:

$$\begin{aligned} \max_P U_{S_U}(P, \lambda) &= (1 - p_s p_a(n)) \sum_{i \in \mathcal{I}(t)} r_{i,k} \sum_{i' \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})] \\ &\quad - \chi p_s p_a(n) \sum_{j \in \mathcal{J}} r_{j,k} - e^{\theta P}, \\ \text{s.t. } 0 &\leq P \leq P_{\max}. \end{aligned} \quad (23)$$

The optimization problem for the attacker is to choose the optimal attack intensity to maximize its utility. Therefore, the optimization problem for the attacker can be formulated as *Problem 2*:

$$\begin{aligned} \max_{\lambda} U_{S_A}(P, \lambda) &= \zeta p_s p_a \sum_{j \in \mathcal{J}} r_{j,k} - p_s p_a \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})], \\ \text{s.t. } 0 &\leq \lambda \leq \lambda_{\max}. \end{aligned} \quad (24)$$

The objective of the Stackelberg game is to find the Stackelberg equilibrium to maximize the utility of each player. The equilibrium solution of the game is defined as follows.

Definition 2: Let P^* be a solution for Problem 1, and λ^* be a solution for Problem 2. $\{P^*, \lambda^*\}$ is a Stackelberg equilibrium for the proposed game if the following conditions are satisfied:

$$U_{S_U}(P^*, \lambda^*) \geq U_{S_U}(P, \lambda^*), \quad (25)$$

$$U_{S_A}(P^*, \lambda^*) \geq U_{S_A}(P^*, \lambda). \quad (26)$$

C. Game Analysis

We choose to use the backward induction method for game analysis rather than the Lagrange method for the following reasons. First, the constraint in Problems 1 and 2 is a simple range-of-values constraint that aims to ensure that the optimal strategy does not exceed the maximum value of the strategy, so the set of available strategies for each player is finite. Therefore, the backward induction method can be directly applied to the Stackelberg game model under numerical constraints. Second, considering that the optimization problem in our paper involves simple numerical constraints instead of complex equations or inequality constraints, the backward induction method presents a more direct and practical approach to facilitate the determination of optimal strategies for the participants in comparison to the Lagrangian method. Therefore, the backward induction method [43] is exploited to obtain the Stackelberg equilibrium by analyzing the strategy of each player. In the backward induction method, each follower is first analyzed and then the leader's strategy is derived. Thus, we first analyze the attacker's decision-making process to obtain the optimal attack intensity strategy in the second stage. Then, we investigate the optimal interference power strategy of the leader UAV in the first stage.

In the second stage: The optimal strategy of the attacker can be obtained by solving Problem 1. The first-order differential of $U_{S_A}(P, \lambda)$ with respect to λ is

$$\frac{\partial U_{S_A}(\lambda)}{\partial \lambda} = (\tilde{\Delta}_1 - \tilde{\Delta}_2) \pi R_a^2 e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^{n-1}}{n!} (n - \lambda \pi R_a^2), \quad (27)$$

where $\tilde{\Delta}_1 = \zeta p_s \sum_{j \in \mathcal{J}} r_{j,k}$, $\tilde{\Delta}_2 = p_s \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})]$.

Then, the second-order differential of the $U_{S_A}(P, \lambda)$ with respect to λ is

$$\begin{aligned} \frac{\partial^2 U_{S_A}(\lambda)}{\partial \lambda^2} &= - \frac{(\tilde{\Delta}_1 - \tilde{\Delta}_2) e^{-\lambda \pi R_a^2} \lambda^{n-2}}{(n-1)!} (\pi R_a^2)^n (\lambda \pi R_a^2 \\ &\quad - n - 1). \end{aligned} \quad (28)$$

Based on (28), if $\lambda \pi R_a^2 - n - 1$ is always positive, $\frac{\partial^2 U_{S_A}(\lambda)}{\partial \lambda^2} < 0$ holds. Thus, the first-order differential of attacker's utility $\frac{\partial U_{S_A}(\lambda)}{\partial \lambda}$ is a strictly monotonic decreasing function. Let

$$\begin{cases} \lim_{\lambda \rightarrow \frac{n-1}{\pi R_a^2}} \frac{\partial U_{S_U}(\lambda)}{\partial \lambda} = (\tilde{\Delta}_1 - \tilde{\Delta}_2) \pi R_a^2 e^{1-n} \frac{(n-1)^{n-1}}{n!} > 0, \\ \lim_{\lambda \rightarrow \infty} \frac{\partial U_{S_U}(\lambda)}{\partial \lambda} = -\infty < 0. \end{cases} \quad (29)$$

Combining (29) with (28), we find that the first-order differential of the attacker's utility is first larger than zero and then smaller than zero. As such, the utility function $U_{S_A}(P, \lambda)$ first increases and then decreases with the increase of λ , that is, $U_{S_A}(P, \lambda)$ is a concave function. Thus, the optimal strategy of the attacker can be obtained by letting $\frac{\partial U_{S_U}(\lambda)}{\partial \lambda} = 0$. In addition, since the objective function of optimization Problem 1 is a concave function, the optimal strategy is equal to the maximum

value of the strategy when the optimal strategy is greater than the maximum value of the strategy. Thus, the optimal strategy of the attacker is

$$\lambda^* = \begin{cases} \frac{n}{\pi R_a^2}, & \text{if } \frac{n}{\pi R_a^2} < \lambda_{\max}, \\ \lambda_{\max}, & \text{otherwise.} \end{cases} \quad (30)$$

In the first stage: Based on the attacker's strategies, we then analyze the leader UAV's strategy. The optimal strategy of leader UAV can be obtained by analyzing Problem 2. Based on the optimal attack intensity of attacker in (30), the utility of UAV group with $\lambda^* = \frac{n}{\pi R_a^2}$ can be rewritten as

$$\begin{aligned} U_{SU} = & \left(1 - p_s e^{-\lambda^* \pi R_a^2} \frac{(\lambda^* \pi R_a^2)^n}{n!}\right) \sum_{i \in \mathcal{I}(t)} r_{i,k} \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \\ & \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{j'})] \\ & - \chi p_s e^{-\lambda^* \pi R_a^2} \frac{(\lambda^* \pi R_a^2)^n}{n!} \sum_{j \in \mathcal{J}} r_{j,k} - e^{\theta P}. \end{aligned} \quad (31)$$

The first-order differential of the leader UAV's utility with respect to P is

$$\frac{\partial U_{SU}(P)}{\partial P} = \tilde{\Delta}_3 \sum_{i \in \mathcal{I}(t)} \frac{1}{1 + P - p_i} - \theta e^{\theta P}, \quad (32)$$

where $\tilde{\Delta}_3 = (1 - p_s e^{-\lambda^* \pi R_a^2} \frac{(\lambda^* \pi R_a^2)^n}{n!}) \sum_{i \in \mathcal{I}(t)} r_{i,k}$.

Since the objective function of optimization Problem 2 is a concave function, as shown in Theorem 2, the optimal strategy is equal to the maximum value of the strategy when the optimal strategy is greater than the maximum value of the strategy. Let $\frac{\partial U_{SU}(P)}{\partial P}$ be equal to zero, the optimal strategy of the leader UAV can be described as

$$P^* = \begin{cases} P_{\max}, & \text{if } \theta^{-1} \ln[(1 - p_s e^{-n \frac{n}{\pi R_a^2}}) \sum_{i \in \mathcal{I}(t)} r_{i,k} + \theta^{-1} \\ & \times (1 - p_s e^{-n \frac{n}{\pi R_a^2}}) c \sum_{i \in \mathcal{I}(t)} r_{i,k}]] > P_{\max}, \\ \theta^{-1} \ln[(1 - p_s e^{-n \frac{n}{\pi R_a^2}}) \sum_{i \in \mathcal{I}(t)} r_{i,k} + \theta^{-1} (1 - p_s e^{-n} \\ & \times \frac{n}{\pi R_a^2}) c \sum_{i \in \mathcal{I}(t)} r_{i,k}]], & \text{otherwise.} \end{cases} \quad (33)$$

where c is an arbitrary constant.

Theorem 2: There only exists one Stackelberg equilibrium for the leader UAV in this Stackelberg game, denoted as $\{P^*, \lambda^*\}$.

Proof: The second-order differential of the UAV group's utility with respect to P is

$$\frac{\partial^2 U_{SU}(P)}{\partial P^2} = -\tilde{\Delta}_3 \sum_{i \in \mathcal{I}(t)} \frac{1}{(1 + P - p_i)^2} - \theta^2 e^{\theta P} < 0. \quad (34)$$

Since $\frac{\partial^2 U_{SU}(P)}{\partial P^2}$ is smaller than zero, the first-order differential of the leader UAV's utility $\frac{\partial U_{SU}(P)}{\partial P}$ is a strictly monotonic

decreasing function. Let

$$\begin{cases} \lim_{P \rightarrow 0} \frac{\partial U_{SU}(P)}{\partial P} = \sum_{i \in \mathcal{I}(t)} \frac{\tilde{\Delta}_3}{1 - p_i} - \theta > 0, \\ \lim_{P \rightarrow \infty} \frac{\partial U_{SU}(P)}{\partial P} = -\infty < 0. \end{cases} \quad (35)$$

Based on (34), the utility function of UAV $U_{SU}(P, \lambda)$ is a strictly concave function. According to the Fermat's theorem [43], there only exists one optimal strategy to maximize the utility of the UAV group. Theorem 2 is proved.

With the above game analysis, the Stackelberg equilibrium between the attacker and the leader UAV can be described as

$$\begin{aligned} (P^*, \lambda^*) = & \left(\theta^{-1} \ln \left[\left(1 - p_s e^{-n \frac{n}{\pi R_a^2}}\right) \sum_{i \in \mathcal{I}(t)} r_{i,k} \right. \right. \\ & \left. \left. + \theta^{-1} \left(1 - p_s e^{-n \frac{n}{\pi R_a^2}}\right) \sum_{i \in \mathcal{I}(t)} r_{i,k} c \right] \right), \frac{n}{\pi R_a^2} \end{aligned} \quad (36)$$

VI. SECURE UAV COMMUNICATION SCHEME BASED ON CUMULATIVE PROSPECT THEORY

As discussed in Section V, the Stackelberg game is utilized to model the decision-making process of the leader UAV and the attacker when the participants are fully rational and the decision is not influenced by any subjective perception. However, as has been shown in many psychological and behavioral studies, individual behaviors can significantly deviate from the completely rational principle of game theory under conditions of risk and uncertainty [44]. Thus, to capture the potential subjective perception (i.e., bounded rationality) of the leader UAV and the attacker, PT is used in the Stackelberg game. In this paper, Stackelberg game theory is a special case of PT, i.e., when the ideal coefficient of participants in PT is set to 1 (i.e., completely rational) and only pure strategy interactions are considered, then PT is equivalent to the Stackelberg game. In fact, using PT, the probability of the utility function and the subjective perception of the utility function relative to the reference point will be central to the decision-making process of the leader UAV and the attacker.

In our model, the uncertainty of the proposed scheme depends on the number of eavesdropping UAVs. Thus, when the leader UAV chooses P and the attacker chooses λ , and if we let n be the number of eavesdropping UAVs selected by the attacker, the possible utility of attacker $U'_{SA}(1, 2, \dots, n, \dots, N)$, the possible utility of UAV group $U'_{SU}(1, 2, \dots, n, \dots, N)$, and the probability of occurrence $\tau(1, 2, \dots, n, \dots, N)$, can be separately defined as

$$\begin{aligned} U'_{SU}(n) = & \left(1 - p_s e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!}\right) \sum_{i \in \mathcal{I}(t)} r_{i,k} \\ & \times \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta\beta}{M-1} \\ & \times \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})] - \chi p_s e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} \sum_{j \in \mathcal{J}} r_{j,k} - e^{\theta P}, \end{aligned} \quad (37)$$

$$\begin{aligned}
U'_{SA}(n) &= \zeta p_s e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} \sum_{j \in \mathcal{J}} r_{j,k} - p_s \\
&\times e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) \\
&+ \log(1 + P - p_i) + \frac{\delta \beta}{M - 1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})], \quad (38)
\end{aligned}$$

$$\tau(n) = e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!}, \quad (39)$$

Here, N is the total number of eavesdropping UAVs.

The purpose of each player in the PT-based game is to maximize its prospect valuation, denoted as $v(U'(n))$, which is defined as

$$v(U'(n)) = \begin{cases} (U'(n) - R_S)^\xi, & U'(n) \geq R_S, \\ -\tilde{\lambda}_S(-(U'(n) - R_S))^\varpi, & U'(n) < R_S, \end{cases} \quad (40)$$

where ϖ and ξ represent the risk appetite degree of players for gains and losses, respectively. $\tilde{\lambda}_S$ is the loss aversion coefficient of players. R_S represents the reference point of players.

Based on the possible outcomes of players and the corresponding probabilities, the $\{P, \lambda\}$ strategy pair produces a prospect Π , in which the outcomes are sorted in ascending order, and is defined as

$$\Pi = \{U'_{SU}(1), U'_{SA}(1), \tau(1); \dots; U'_{SU}(n), U'_{SA}(n), \tau(n); \dots; U'_{SU}(N), U'_{SA}(N), \tau(N)\}. \quad (41)$$

According to the sign of $v(U'(n))$, the prospect Π can be divided into positive prospect Π^+ and negative prospect Π^- , which are calculated as [44]

$$V(\Pi^+) = \sum_{i=N_a+1}^N \pi_i^+ v(U'(i)), \quad (42)$$

$$V(\Pi^-) = \sum_{i=1}^{N_a} \pi_i^- v(U'(i)), \quad (43)$$

where N_a is the index of the middle value among all outcomes sorted in ascending order. As such, the valuation of prospect Π is $V(\Pi)$, $V(\Pi) = V(\Pi^+) + V(\Pi^-)$. π_i^+ and π_i^- are probability weight functions, related to the probability of each possible outcome, given by [44]

$$\pi_i^+ = \omega^+ \left(\sum_{j=i}^N \tau(j) \right) - \omega^+ \left(\sum_{j=i+1}^N \tau(j) \right), \quad (44)$$

$$\pi_i^- = \omega^- \left(\sum_{j=1}^i \tau(j) \right) - \omega^- \left(\sum_{j=1}^{i-1} \tau(j) \right), \quad (45)$$

where ω^+ and ω^- are the decision weight functions, which represent the player's subjective evaluation of objective probability. Based on the objective probability $\tau(i)$, the decision weight

functions ω^+ and ω^- can be calculated separately as

$$\omega^+(\tau) = \frac{\tau^{\gamma^+}}{(\tau^{\gamma^+} + (1 - \tau)^{\gamma^+})^{1/\gamma^+}}, \quad (46)$$

$$\omega^-(\tau) = \frac{\tau^{\gamma^-}}{(\tau^{\gamma^-} + (1 - \tau)^{\gamma^-})^{1/\gamma^-}}, \quad (47)$$

where γ^+ and γ^- are known rational parameters.

Using PT, the attacker and the leader UAV subjectively evaluate every possible outcome of prospect Π . Therefore, the leader UAV's subjective evaluation of the n th possible outcome is $v(U'_{SU}(n))$, which is defined as

$$v(U'_{SU}(n)) = \begin{cases} (\Delta \mathcal{C}_n)^{\xi_{SU}}, & \Delta \mathcal{C}_n \geq 0, \\ -\tilde{\lambda}_{SU}(-(\Delta \mathcal{C}_n))^{\varpi_{SU}}, & \Delta \mathcal{C}_n < 0, \end{cases} \quad (48)$$

where

$$\begin{aligned}
\Delta \mathcal{C}_n &= \left(1 - p_s e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} \right) \sum_{i \in \mathcal{I}(t)} r_{i,k} \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \\
&\times \log(1 + p_i) + \log(1 + P - p_i) + \frac{\delta \beta}{M - 1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'})] \\
&- \chi p_s e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} \sum_{j \in \mathcal{J}} r_{j,k} - e^{\theta P} - R_{SU}. \quad (49)
\end{aligned}$$

Since the goal of the leader UAV is to maximize its utility, $\Delta \mathcal{C}_n \geq 0$ is considered as gain, while $\Delta \mathcal{C}_n < 0$ is considered as loss.

Similarly, the attacker's subjective evaluation of the n th possible outcome is $v(U'_{SA}(n))$, which is described as

$$v(U'_{SA}(n)) = \begin{cases} (\Delta \mathcal{N}_n)^{\xi_{SA}}, & \Delta \mathcal{N}_n \geq 0, \\ -\tilde{\lambda}_{SA}(-(\Delta \mathcal{N}_n))^{\varpi_{SA}}, & \Delta \mathcal{N}_n < 0, \end{cases} \quad (50)$$

where

$$\begin{aligned}
\Delta \mathcal{N}_n &= \zeta e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} p_s \sum_{j \in \mathcal{J}} r_{j,k} - e^{-\lambda \pi R_a^2} \frac{(\lambda \pi R_a^2)^n}{n!} p_s \\
&\times \sum_{i \in \mathcal{I}(t)} [(1 - \delta) \log(1 + p_i) + \log(1 + P - p_i)] \\
&+ \frac{\delta \beta}{M - 1} \sum_{i' \in \mathcal{I}(t) \setminus i} \log(1 + p_{i'}) - R_{SA}. \quad (51)
\end{aligned}$$

Given that the purpose of the attacker is to maximize its utility, $\Delta \mathcal{N}_n \geq 0$ is considered as gain, while $\Delta \mathcal{N}_n < 0$ is considered as loss.

Based on the PT principles, we can obtain the valuations that S_A and S_U assign to each possible outcome of strategy $\{P, \lambda\}$.

Theorem 3: The PT valuation that S_A assigns to a strategy pair $\{P, \lambda\}$ is defined as

$$V_{S_A}(\Pi) = \sum_{i=N_a+1}^{\infty} (\Delta \mathcal{N}_i)^{\xi_{S_A}} \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right] + \sum_{i=0}^{N_a} (-\tilde{\lambda}_{S_A}(-(\Delta \mathcal{N}_i))^{\varpi_{S_A}}) \times \left[\omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^i \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^{i-1} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right], \quad (52)$$

where $\Delta \mathcal{N}_n > 0$, if $n \geq N_a + 1$. $\Delta \mathcal{N}_n < 0$, if $n \leq N_a$.

Proof: When the attacker assigns to a strategy pair $\{P, \lambda\}$, the attacker's PT valuation is defined as

$$V_{S_A}(\Pi) = V_{S_A}(\Pi^+) + V_{S_A}(\Pi^-) \\ = \sum_{i=N_a+1}^{\infty} \pi_i^+ v(U'_{S_A}(i)) + \sum_{i=0}^{N_a} \pi_i^- v(U'_{S_A}(i)), \quad (53)$$

where

$$\sum_{i=0}^{N_a} \pi_i^- v(U'_{S_A}(i)) = \Psi(0)[\omega^-(p_a(0))] + \Psi(1)[\omega^-(p_a(0) + p_a(1)) - \omega^-(p_a(0))] + \cdots + \Psi(N_a)[\omega^-(p_a(0) + \cdots + p_a(N_a)) - \omega^-(p_a(0) + \cdots + p_a(N_a - 1))] \\ = \sum_{i=0}^{N_a} \Psi(i) \left[\omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^i \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^{i-1} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right], \quad (54)$$

where $\Psi(i) = (-\tilde{\lambda}_{S_A}(-(\Delta \mathcal{N}_i))^{\varpi_{S_A}})$.

$$\sum_{i=N_a+1}^{\infty} \pi_i^+ v(U'_{S_A}(i)) = (\Delta \mathcal{N}_{N_a+1})^{\xi_{S_A}} [\omega^+(p_a(N_a + 1) + \cdots + p_a(\infty)) - \omega^+(p_a(N_a + 2) + \cdots + p_a(\infty))] \\ + (\Delta \mathcal{N}_{N_a+2})^{\xi_{S_A}} [\omega^+(p_a(N_a + 2) + \cdots + p_a(\infty)) - \omega^+(p_a(N_a + 3) + \cdots + p_a(\infty))] + \cdots \\ = \sum_{i=N_a+1}^{\infty} (\Delta \mathcal{N}_i)^{\xi_{S_A}} \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right]. \quad (55)$$

Therefore, Theorem 3 is proved.

Theorem 4: The PT valuation that S_U assigns to a strategy pair $\{P, \lambda\}$ is defined as

$$V_{S_U}(\Pi) = \sum_{i=N_a+1}^{\infty} (\Delta \mathcal{C}_i)^{\xi_{S_U}} \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=i+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right] + \sum_{i=0}^{N_a} (-\tilde{\lambda}_{S_U}(-(\Delta \mathcal{C}_i))^{\varpi_{S_U}}) \times \left[\omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^i \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^- \left(e^{-\lambda \pi R_a^2} \sum_{j=0}^{i-1} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right], \quad (56)$$

if $n \geq N_a + 1$, $\Delta \mathcal{C}_n > 0$. If $n \leq N_a$, $\Delta \mathcal{C}_n < 0$.

Proof: The steps of the proof are similar to that of Theorem 3.

Based on (52) and (56), we know that $V_{S_U}(\Pi)$ and $V_{S_A}(\Pi)$ are the sum of infinite series. Thus, to be able to compare the PT valuations of different strategies for $\{P, \lambda\}$, we need to prove the convergence of $V_{S_A}(\Pi)$ and $V_{S_U}(\Pi)$.

Theorem 5: When $\gamma^+ = 1$, $V_{S_A}(\Pi)$ is a convergent series.

Proof: To prove the convergence of $V_{S_A}(\Pi)$, we first prove the convergence of $V_{S_A}(\Pi^+)$. Since $V_{S_A}(\Pi^+)$ is defined in (55) and composed of positive terms, we use the ratio test to prove its convergence. Let $V_{S_A}^n(\Pi^+)$ be the n th item in $V_{S_A}(\Pi^+)$, based on the definition in (55), $V_{S_A}(\Pi^+)$ is expressed as

$$V_{S_A}^n(\Pi^+) = (\Delta \mathcal{N}_n)^{\xi_{S_A}} \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right]. \quad (57)$$

Therefore, following the ratio test, we have

$$\tilde{L} = \lim_{n \rightarrow \infty} \frac{V_{S_A}^{n+1}(\Pi^+)}{V_{S_A}^n(\Pi^+)} \\ = \left\{ (\Delta \mathcal{N}_{n+1})^{\xi_{S_A}} \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n+2}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right] \right\} / \left\{ (\Delta \mathcal{N}_n)^{\xi_{S_A}} \right. \\ \times \left[\omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) - \omega^+ \left(e^{-\lambda \pi R_a^2} \sum_{j=n+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} \right) \right] \right\}. \quad (58)$$

Let $T(x)$ be the Taylor expansion of e^x , as such, $\sum_{j=n+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!}$ can be rewritten as

$$\sum_{j=n+1}^{\infty} \frac{(\lambda \pi R_a^2)^j}{j!} = T(\lambda \pi R_a^2) - T_n(\lambda \pi R_a^2), \quad (59)$$

where T_n is the previous n term of $T(x)$.

Thus, we have

$$\tilde{L} = \lim_{n \rightarrow \infty} \frac{V_{S_A}^{n+1}(\Pi^+)}{V_{S_A}^n(\Pi^+)} = \lim_{n \rightarrow \infty} \frac{\Xi_n - \Xi_{n+1}}{\Xi_{n-1} - \Xi_n} = 0 < 1, \quad (60)$$

where

$$\begin{aligned} \Xi_n &= [T(\lambda\pi R_a^2) - T_n(\lambda\pi R_a^2)]^{\gamma^+}, \\ \Xi_{n+1} &= [T(\lambda\pi R_a^2) - T_{n+1}(\lambda\pi R_a^2)]^{\gamma^+}, \\ \Xi_{n-1} &= [T(\lambda\pi R_a^2) - T_{n-1}(\lambda\pi R_a^2)]^{\gamma^+}. \end{aligned} \quad (61)$$

Theorem 5 is proved.

Theorem 6: When $\gamma^+ = 1$, $V_{S_U}(\Pi)$ is a convergent series.

Proof: The steps of the proof are similar to that of Theorem 5.

The complexity analysis of the proposed scheme consists of two parts. The first part is from obtaining the optimal strategies for $M - 1$ legal UAVs. The optimal strategy for each legal UAVe is whether or not to join the grand coalition. Specifically, each legal UAV chooses whether or not to join the grand coalition by calculating the utility when joining the coalition and when not joining the coalition. In addition, to defend against eavesdropping attacks, all UAVs within the coalition work as a whole, with the leader UAV choosing the interference power strategy to maximize the utility of the UAV group. The analysis of the optimal strategy for each UAV is beyond the scope of this paper, and we can investigate this aspect in the future. Therefore, the complexity of the first part is $\mathcal{O}(M - 1)$. The second part is divided into two cases. Case 1 is from searching for the optimal strategy for the attacker and the leader UAV when the participants are fully rational. Case 2 is from searching for the optimal strategy for the attacker and the leader UAV when the participants are bounded rational. Let I_t denote the number of iterations in the second part. Therefore, the complexity of the first part is $\mathcal{O}(I_t \times N)$. Thus, the total complexity of the proposed scheme is $\mathcal{O}((M - 1) + I_t \times N)$.

VII. PERFORMANCE EVALUATION

In this section, we evaluate the performance of the proposed scheme by using simulators operated on Matlab. The simulation setup is first introduced, followed by the numerical results and analysis.

A. Simulation Setup

In the simulation scenario, although the number of legal UAVs has a potential impact on the security of the UAV networks, the issue regarding the scheduling of legal UAVs is outside the scope of this paper. Due to the time-varying decision-making process of legal UAVs and the mobility of both legal UAVs and eavesdropping UAVs, the decision-making can be done in each time slot. We can directly analyze the decision-making process during one time slot. Furthermore, in the mobility model, since ϱ can be adequately small, the position of the UAV can be approximately fixed in each time slot. Therefore, the mobility of the UAV has no impact on the performance of the proposed scheme. Meanwhile, all legal UAVs form a grand coalition to jointly resist eavesdropping attacks before communicating with ground

TABLE II
SIMULATION PARAMETERS

Parameters	Values
B	1MHz
θ	[0, 1]
R_u	158m
R_a	100m
p_i	[10, 40]dBm
$\mathbf{p}_i$	[0, 10]dBm
δ	0.6
β	0.3
$\tilde{\lambda}_{S_U}$	[0.5, 0.8]

nodes, the number of legal UAVs in the grand coalition is chosen randomly in [20, 25] [45]. In each time slot, we select the legal UAV with the maximum interference power as the leader UAV based on the interference power magnitude of each legal UAV. In addition, in the Stackelberg game, a purely strategic interaction occurs between the attacker and the leader UAV, so the number of eavesdropper UAVs is required to be determined. On the other hand, in PT, the uncertainty of the proposed scheme depends on the number of eavesdropping UAVs, and thus the number of eavesdropping UAVs is constantly changing in the experiment. Therefore, the number of eavesdropping UAVs is chosen in [1, 20] at random. The flying altitudes of eavesdropping UAVs and legal UAVs are fixed at 120 m and 100 m, respectively [35]. In addition, the noise power σ^2 is set to -104 dB [46], and the reference channel power h is set to -50 dB [47]. The probability of a successful attack is randomly picked within [0.1, 1]. The distance d_u between the airspace of the legal UAVs Ω_u and the airspace of eavesdropping UAVs Ω_e is set to 20 m [38]. The maximum flight speed of legal UAV i is set to 30 m/s [7]. The number of iterations in the experiments is 100. Other parameters in the simulation are listed in Table II [3], [4].

To show the superiority of our proposed scheme, the following two conventional schemes are utilized for comparisons.

- *The random Scheme* [48]: In the random scheme, the attacker chooses a random attack intensity strategy and the leader UAV randomly selects an interference power strategy.
- *The fixed Scheme* [14]: In the fixed scheme, the leader UAV selects a fixed interference power strategy. The attacker randomly chooses the attack intensity strategy.

B. Numerical Results and Analysis

Fig. 2 shows the performance of the utility for the legal UAV with different p_i . To demonstrate the decision-making process of the legal UAV, we assume that the attack intensity of the attacker λ is set to 1. From Fig. 2, we can observe that as the interference power of the legal UAV increases, the utility of the legal UAV that joins the coalition first increases and then decreases. The reason is that the goal of the proposed scheme is to maximize the number of legal UAVs participating in the coalition. Thus, when legal UAVs choose to participate in the coalition, the strictly concave function is used to the utility of the legal UAV for security. Instead of allocating more defensive resources to legal UAVs in the coalition to achieve a higher

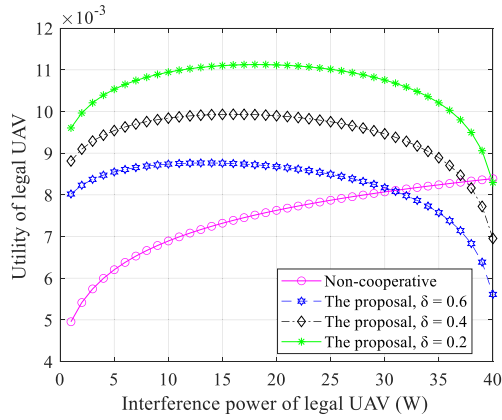


Fig. 2. Impacts of interference power p_i on the utility of legal UAV.

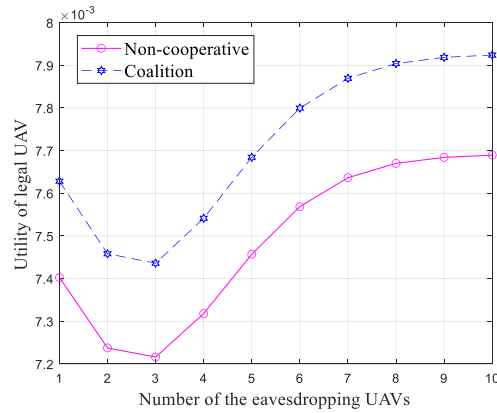


Fig. 3. Changes of legal UAV utility with the different numbers of eavesdropping UAVs.

secrecy rate, the leader UAV allocates its defensive resources guided by the concave utility function to motivate the highest possible number of legal UAVs participating in the coalition. Besides, we can see that the utility of the legal UAV that joins the coalition is first greater than the utility of the legal UAV without joining the coalition, and then less than the utility of the legal UAV without joining the coalition. This is because the defense capability of the legal UAV is positively related to the interference power of the legal UAV when opting out of a coalition. Moreover, when p_i increases to not satisfy $P \gg p_i$, the optimal strategy of each legal UAV is not to join the coalition, i.e., the utility of the legal UAV that does not participate in the coalition is greater than the utility of the legal UAV that does participate in the coalition. Furthermore, the utility of the legal UAV increases as the defense resources decrease. The reason is that the fewer defense resources a legal UAV shares, the more defense resources it can use to defend against eavesdropping attacks with the same proportion of resources shared by other UAVs, and thus the higher its utility.

The variation of the legal UAV's utility with different numbers of eavesdropping UAVs is shown in Fig. 3. In this simulation, the interference power of legal UAV p_i is set to 20 and the interference power of the leader UAV is set to 40. From Fig. 3,

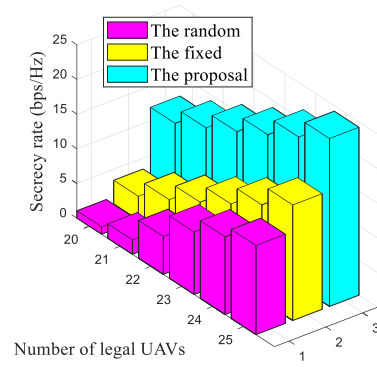


Fig. 4. Comparison of the proposed scheme with conventional schemes on the secrecy rate of UAV networks, when the number of legal UAVs changes.

we can observe that as the number of eavesdropping UAVs increases, the utility of the legal UAV first decreases and then increases. The reason is that the attack probability of the attacker is subject to the Poisson process, which means that the attack probability of the attacker first increases and then decreases. In addition, we can see that under the satisfaction of the power constraint, regardless of the number of eavesdropping UAVs, the utility of the legal UAV is greater when it chooses to join the coalition than when it does not. Note that legal UAVs are selfish, i.e., each legal UAV will choose the optimal strategy to maximize its utility. Therefore, legal UAVs are motivated to join the coalition to collaboratively defend against eavesdropping attacks.

The secrecy rate is the primary design metric for physical-layer security, which is expressed as the ratio of confidential information that can be reliably transmitted without being eavesdropped on. Therefore, we measure the security performance of UAV networks in terms of secrecy rate, which is defined as the available transmission rate between the legal UAV and the ground node minus the available transmission rate between the eavesdropping UAV and the ground node.

Fig. 4 shows the performance of UAV networks' secrecy rate in terms of the number of legal UAVs. From Fig. 4, we can see that the secrecy rate of all schemes increases with the number of legal UAVs, and the proposed scheme can achieve the maximum secrecy rate. The reason is that the optimal interference power strategy of the leader UAV can be obtained by utilizing the Stackelberg game and therefore the UAV network has the maximum secrecy rate. In the fixed scheme, the leader UAV chooses the fixed interference power without considering the interaction with the attacker, where the secrecy rate is not maximum. In the random scheme, the leader UAV chooses the strategy randomly, which results in a low secrecy rate.

Fig. 5 shows the performance of the secrecy rate of UAV networks in terms of the number of eavesdropping UAVs. From Fig. 5, we can see that the secrecy rate of UAV networks first decreases and then increases as the number of eavesdropping UAVs increases. This is because the attack probability of the attacker first increases and then decreases, which obeys the Poisson process. Furthermore, in this paper, we consider that the leader UAV will resist eavesdropping attacks by emitting

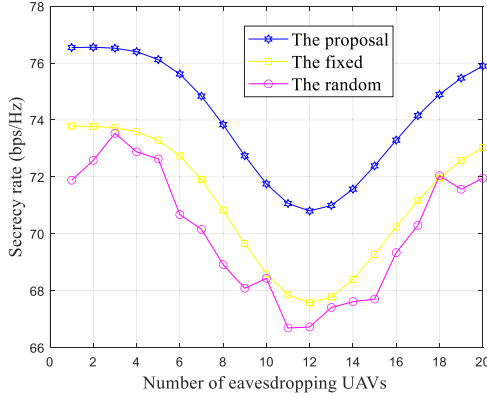


Fig. 5. Comparison of the proposed scheme with conventional schemes on the secrecy rate of UAV networks, when the number of eavesdropping UAVs changes.

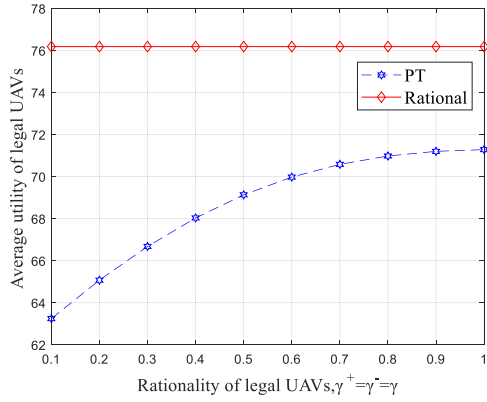


Fig. 6. Average utility of the UAV group for different rationality $\gamma = \gamma^+ = \gamma^-$.

jamming signal, so the attacker is not guaranteed to succeed in every attack. Therefore, the data transmission rate between the eavesdropping UAV and the ground node is positively correlated with the probability that the eavesdropping attack is successful, so the secrecy rate of UAV networks is related to the probability that the eavesdropping attack is successful. Moreover, we can observe that the proposed scheme obtains a higher secrecy rate of UAV networks than the other two schemes. Since the experimental scenario considered in this paper is a small-scale UAV network, there is no very significant difference in the secrecy rate of the UAV networks compared to the benchmarks, and this difference will become larger when the simulation scale is expanded.

Fig. 6 shows the performance of the average utility of the UAV group with different rationality, where the degree of risk appetite of legal UAVs for profits is set to 0.25, the degree of risk appetite of legal UAVs for loss is set to 0.1, and the loss aversion coefficient of legal UAVs is set to 0.8. From Fig. 6, we can observe that the average utility of the UAV group increases with the degree of rationality under PT. The reason is that the rationality of legal UAVs affects their choice of strategy. As the degree of rationality increases, legal UAVs can obtain higher profits. Moreover, we can see that the average utility of the UAV

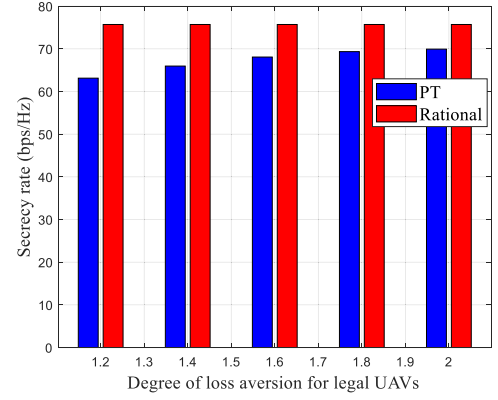


Fig. 7. Average secrecy rate of the legal UAVs for different degree of loss aversion, $\tilde{\lambda}_{S_U}$.

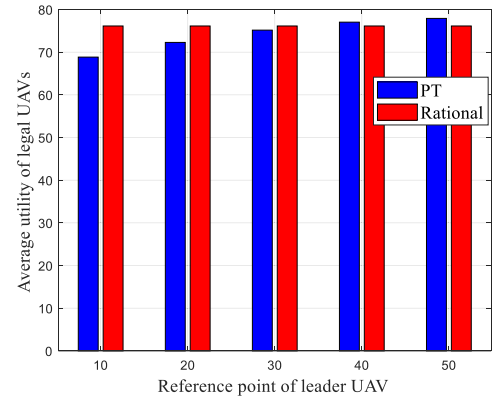


Fig. 8. Average utility of the legal UAVs for different reference point of leader UAV, R_{S_U} .

group under completely rational is better than the average utility of the UAV group under PT. This is because the PT-based game takes into account the underlying rationality of all players, while the Stackelberg game considers all players to be completely rational. The simulation results reflect the player's subjective perception negatively affects the utility of the UAV group.

Fig. 7 shows the variation of the secrecy rate in terms of different levels of loss aversion. From Fig. 7, we can observe that the secrecy rate of UAV networks remains unchanged under complete rational, while the secrecy rate of UAV networks increases with the $\tilde{\lambda}_{S_U}$ under PT. The reason is that the increase of $\tilde{\lambda}_{S_U}$ will cause legal UAVs being more sensitive to losses under PT. Therefore, legal UAVs tend to choose the optimal strategy and the average secrecy rate can be improved. However, in the fully rational case, the decisions of legal UAVs are not affected by the level of loss aversion, so the security rate of UAV networks is unchanged. In addition, the average secrecy rate under complete rationality is better than that under PT, which reflects the impact of players' subjective perception on the average secrecy rate of UAV networks.

Fig. 8 shows the average utility of legal UAVs in terms of different values of the reference point of the leader UAV. From Fig. 8, we can observe that the average utility of legal UAVs under PT is smaller than that of legal UAVs under completely

rational at low values of R_{SU} . However, for relatively high values of R_{SU} , the average utility of legal UAVs under PT is larger than that of legal UAVs under completely rational. The reason is that legal UAVs regard the utilities greater than the values of R_{SU} as gains, and the utilities less than the values of R_{SU} as losses. Besides, UAVs are more sensitive to losses compared to gains. As such, for low values of R_{SU} , legal UAVs adopt a more conservative strategy, while for high values of R_{SU} , the legal UAVs adopt a more risky strategy even without considering the security of UAV networks.

VIII. CONCLUSION

In this paper, we have proposed a framework in which multiple legal UAVs cooperate to transmit data with ground nodes considering the presence of an attacker. In particular, an incentive scheme has been proposed to motivate legal UAVs cooperatively to resist malicious attacks, where the interactions among legal UAVs have been characterized by the coalition game. With the designed scheme, we have proved that the optimal strategy of each legal UAV is to join the coalition to defend against attackers synergistically. Moreover, a secure UAV communication scheme has been proposed to enhance the security of UAV networks, where the interaction between leader UAV and attacker is formulated as a two-stage Stackelberg game in a completely rational scenario. Furthermore, PT has been employed in the above game to model the actual decision-making process of players under conditions of risk and uncertainty. Finally, the simulation results have demonstrated that the proposed scheme enhances the security of UAV networks while improving the utilities of legal UAVs. For future work, we will consider the security issues of UAV networks with multiple attackers.

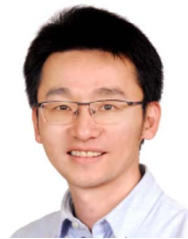
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Liang Xie is currently working toward the PhD degree with the School of Cyber Science and Engineering, Xi'an Jiaotong University, Xi'an, China. His research interests include security and privacy protection in wireless networks and UAV networks.



Zhou Su has published technical papers, including top journals and top conferences, such as *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Mobile Computing*, *IEEE/ACM Transactions on Networking*, and *INFOCOM*. His research interests include multimedia communication, wireless communication, and network traffic. He received the Best Paper Award of International Conference IEEE ICC2020, IEEE Big-dataSE2019, and IEEE CyberSciTech 2017. He is an associate editor of *IEEE Internet of Things Journal*, *IEEE Open Journal of Computer Society*, and *IET Communications*.



Qichao Xu received the PhD degree from the School of Mechatronic Engineering and Automation, Shanghai University, Shanghai, P. R. China, in 2019. He is currently an associate professor with Shanghai university. His research interests are in trust and security, the general area of wireless network architecture, Internet of Things, vehicular networks, and resource allocation.



Nan Chen received the PhD degree in electrical and computer engineering from the University of Waterloo, Canada, in 2019. She is currently an assistant professor with the Department of Electrical and Computer Engineering, Tennessee Tech University. Her research interests include electric vehicle charging/discharging scheme design in smart grid, next-generation wireless networks, and machine learning application in vehicular cyberphysical systems.



Yixin Fan is currently working toward the master's degree with the School of Communications Engineering, Xidian University, Xi'an, China. Her research is dedicated to the integration of communication, sensing, and computing within the vehicular networks domain.



Abderrahim Benslimane is currently a full professor with the Laboratory of Computer Sciences, Avignon University, France. He is also the chair of the ComSoc Technical Committee of Communication and Information Security. He is the EiC of International Journal of Multimedia Intelligence and Security (IJMIS), an area editor of security in *IEEE Internet of Things Journal* and *Security and Privacy Journal* (Wiley), and an editorial member of *IEEE Wireless Communication Magazine*, *Ad Hoc Networks* (Elsevier), *IEEE Systems Journal*, and *IEEE Wireless Communications*.